

W1X1003 Problem Solving in Programming (Tutorial 1)

Part 1

Input	Process	Output
Number 1	Multiply the	Display result of
Number 2	two numbers	multiplication

Pseudocode

Begin

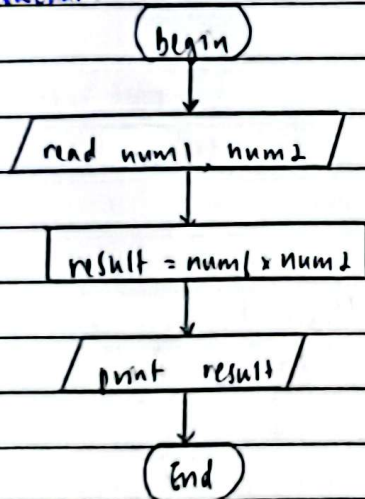
read num1, num2

result = num1 x num2

print result

End

Flowchart



Input	number
Process	Determine message to display based on number
Output	"Number is greater than 50" or "Number is not greater than 50"

Pseudocode

begin

read number

if (number > 50)

print "Number is greater than 50"

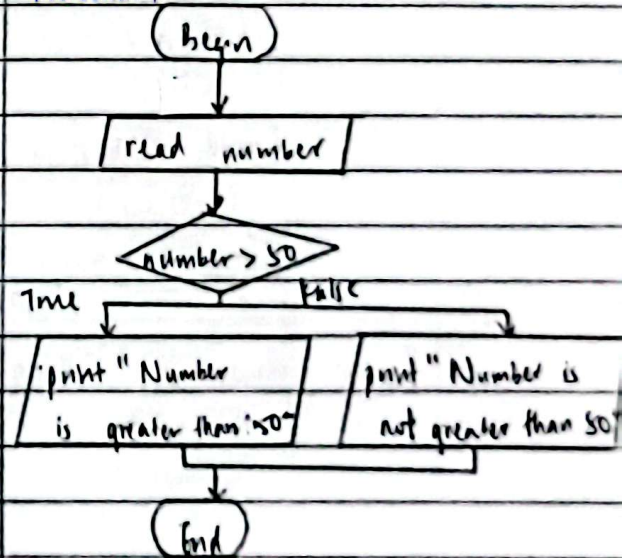
else

print "Number is not greater than 50"

endif

End

Flowchart



3.	Input:	mark
	Process:	Determine message to display based on mark
	Output:	message "Pass grade" or "Fail grade"

Pseudocode

begin

read mark

 if (mark $\geq$ 40)

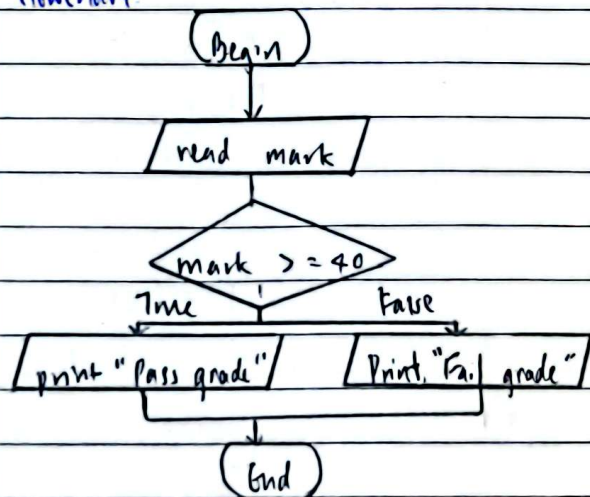
print "Pass grade"

else

print "Fail grade"

end if

End

Flowchart

4.	Input:	player1, player2
	Process:	Determine message to display based on player1 and player2
	Output:	message "Player1 win" or "Player 2 win" or "It's a tie"

Pseudocode

Begin

read player1, player2

if (player1 > player2)

print "Player1 win"

else if (player2 > player1)

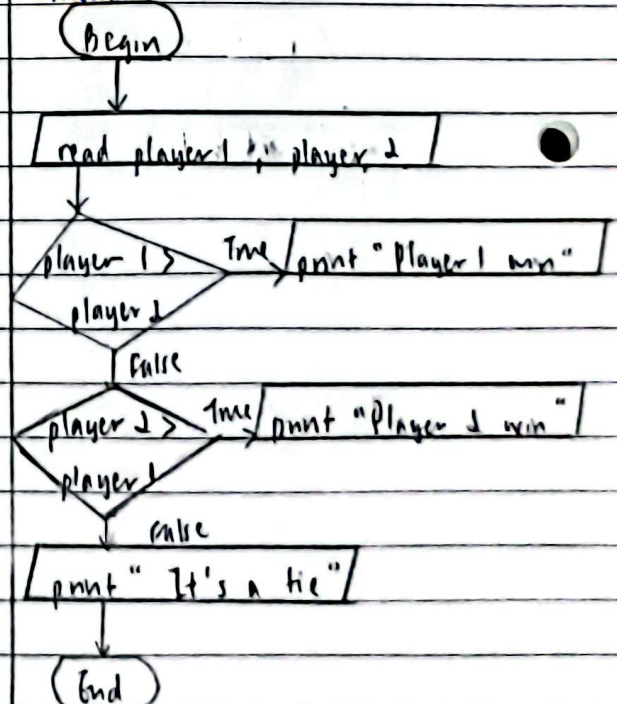
print "Player 2 win"

else

print "It's a tie"

end if

End.

Flowchart

5.

Input	side1, side2, side3
Process	Calculate perimeter of triangle
Output	perimeter

Pseudocode

begin

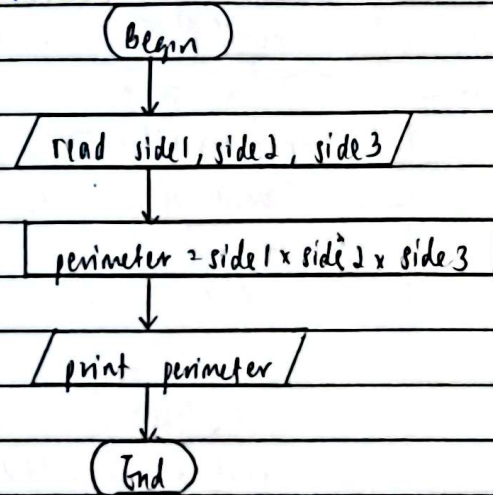
read side1, side2, side3

perimeter = side1 + side2 + side3

print perimeter

end

Flowchart



6.

Input	10 random numbers
Process	Determine minimum number based on 10 numbers
Output	Minimum number

Pseudocode

begin

i = 1

min = 0

while (i <= 10)

read num

if (num < min)

min = num

endif

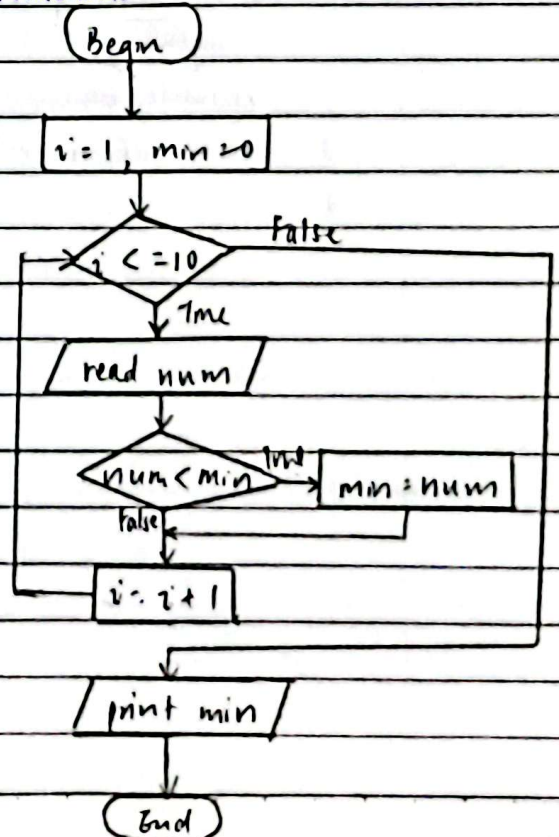
i = i + 1

endwhile

print min

end.

Flowchart.



7.

Input:

-

Process

Determine noOdd and noEven
based on 10 random number
btw 10-100

Output:

noOdd, noEven

Pseudocode

begin

 $i = 1$

noEven = 0

noOdd = 0

min = 10

max = 101

while ($i \leq 10$)

num = randomFunction(min, max)

 if ($\text{num} \% 2 \neq 0$)

noEven++

else

noOdd++

end if

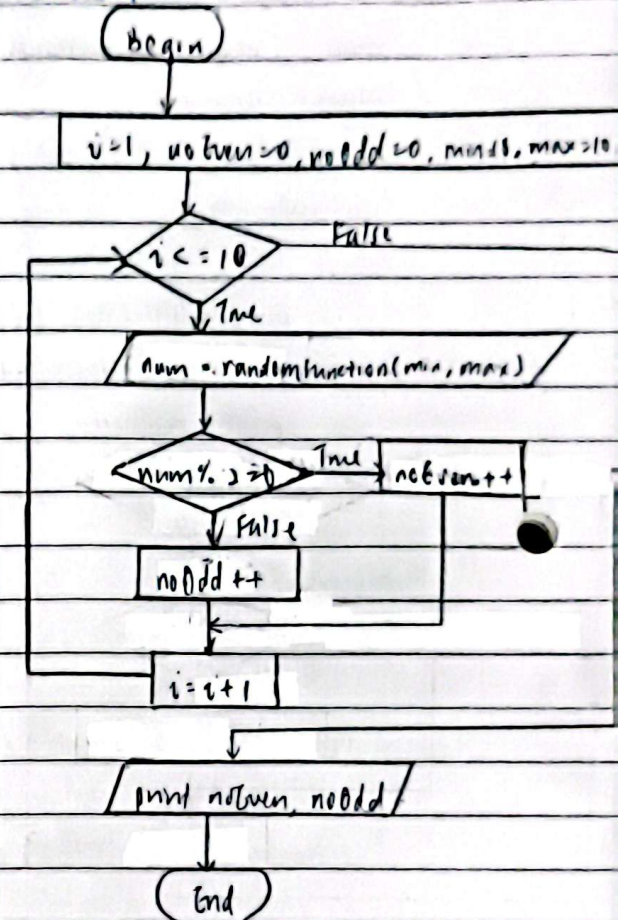
 $i = i + 1$

end while

print noEven, noOdd

End

Flowchart



Part II

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Input: sentence from user

Process: Determine noU and noM based on sentence entered by user

Output: noU, noM

Pseudocode

begin

read sentence

noU ← 0

noM ← 0

while (i < sentence length)

read char

if (char == 'U')

noU ++

else if (char == 'M')

noM ++

endif

i = i + 1

endwhile

print noU, noM

end

Flowchart

Begin

read sentence

noU ← 0, noM ← 0

i < sentence length

read char

char == 'U'

noU ++

char == 'M'

noM ++

i = i + 1

print noU, noM

End

9.	Input	keyword
	Process	Calculate frequency based on keyword entered.
	Output	frequency

Pseudocode

Begin

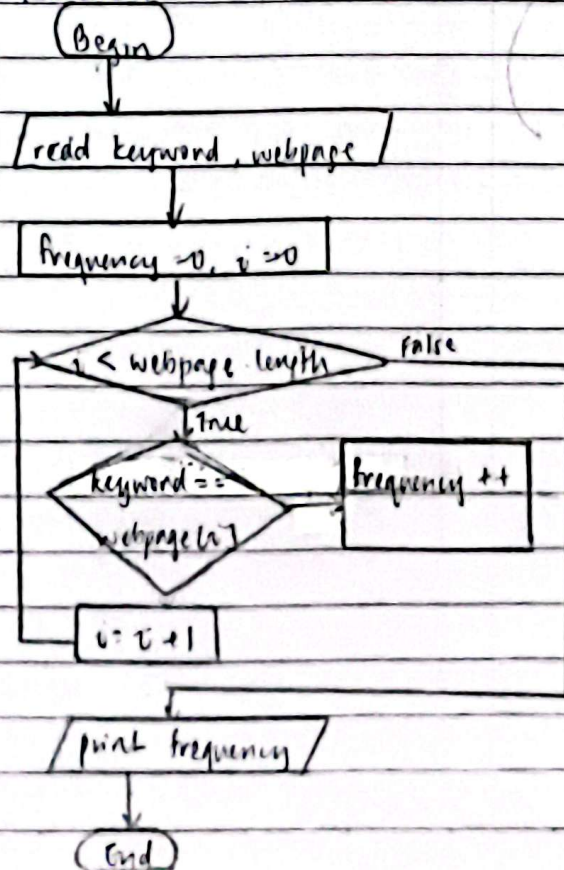
```

read keyword, webpage
frequency = 0, i = 0
while (i < webpage.length)
    if (keyword == webpage[i])
        frequency++
    endif
    i = i + 1
endwhile
print frequency

```

End.

Flowchart



10.	Input	random list of 100 student
	Process	Determine noFemale based on random list of 100 student
	Output	noFemale

Pseudocode

begin

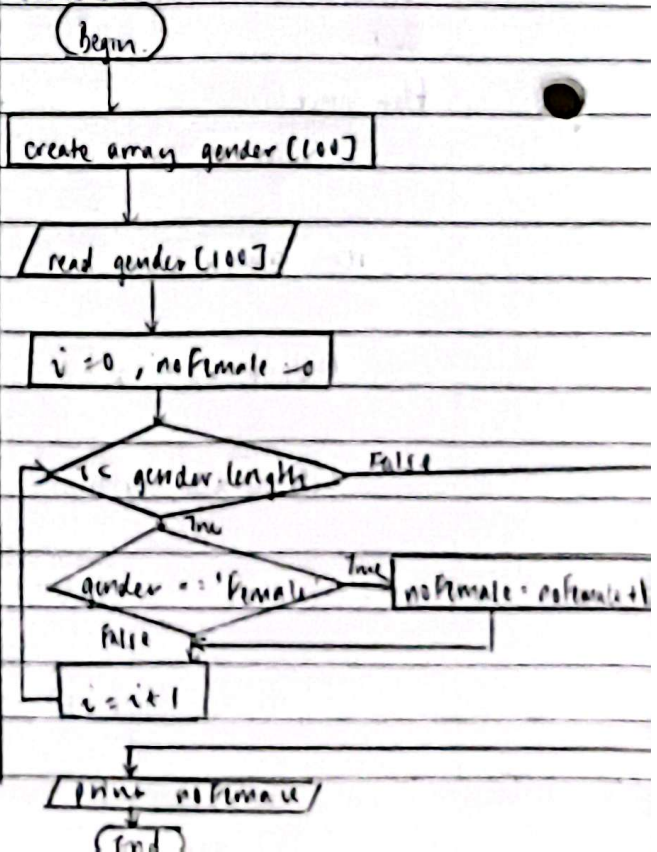
```

create array, gender[100]
read gender[100]
i = 0, noFemale = 0
while (i < gender.length)
    if (gender == 'Female')
        noFemale = noFemale + 1
    endif
    i = i + 1
endwhile
print noFemale

```

End

Flowchart



11

Input	-
Process	Generate 5 random number Sort and display descending order of 5 random number
Output	List of sorted 5 random number

Pseudocode

Begin

create array num[5]

i = 0

while (i < 5)

num[i] = GENERATE_RANDOM_INTEGER()

i = i + 1

endwhile

Arrays.sort(num)

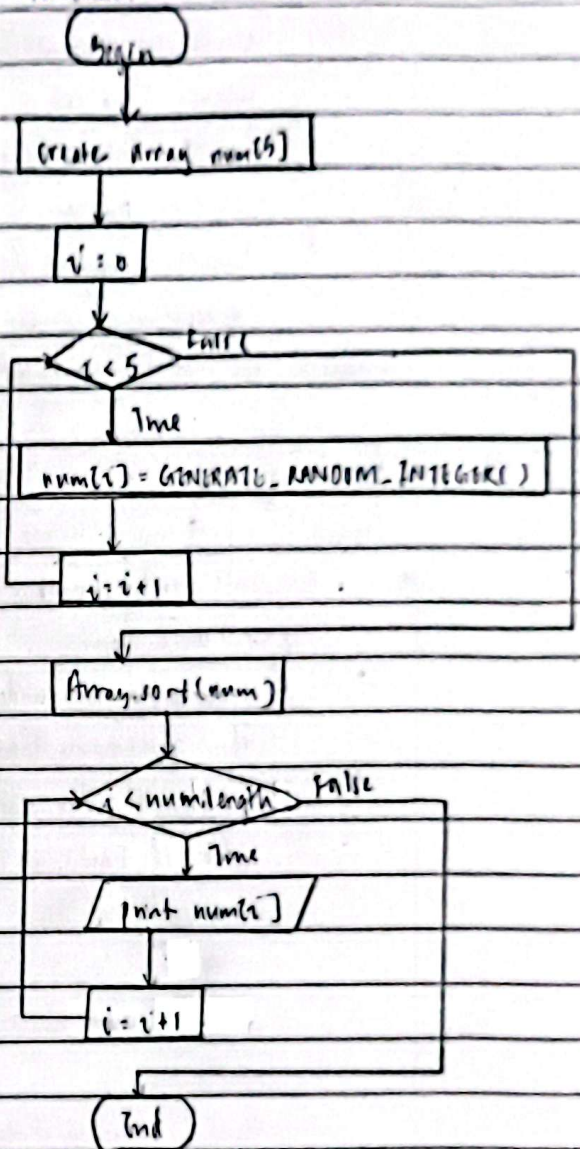
while (i < num.length)

print num[i]

i = i + 1

endwhile

End.

Flowchart

12.

Input	guess
Process	Generate a random number between 1 and 100. Compare user's guess with the random number. Loop until correct number is guessed.
Output	message "Congratulation".

Pseudocode

begin

Generate random-num

read guess-num

 while (guess-num $\neq$ random-num)

if (guess-num > random-num)

print "Too high"

else

print "Too low"

end-if

read guess-num

endwhile

Print "Congratulation"

end

Flowchart